

• E_n : $\langle E \rangle = \sum_n E_n P(n)$

$P(n) = \frac{1}{n}$

• Exam: (2): 70 (3): 80 (1) 90

$$\text{Avg} = \frac{2 \cdot 70 + 3 \cdot 80 + 1 \cdot 90}{6}$$

$$= \left(\frac{2}{6}\right) 70 + \left(\frac{3}{6}\right) 80 + \left(\frac{1}{6}\right) 90$$

* Discrete $\rightarrow$ Continuous
 $\sum_n \rightarrow \int$

$$\langle M \rangle = \int M \rho(M) dM$$

$$= \int M \xi(M) dM$$

$\rho(M) = \xi(M) dM$

$\rho(M) = M \rho(M) = M \xi(M) dM$

$$\langle M \rangle = \int M (M \xi(M) dM)$$

$$= \int \underline{M^2} \underline{\xi(M)} \underline{dM}$$

* $\frac{I_1}{I_2} = 100^{\frac{2}{5}(M_2 - M_1)}$ * *

• $\log_{10} \left(\frac{L_V}{L_{V0}} \right) \leftarrow$ apparent
 • $\log_{10} \left(\frac{L_V}{L_{V0}} \right) \leftarrow$ Absolute

$$I_1 = \frac{L_V}{4\pi D^2}, \quad I_2 = \frac{L_{V0}}{4\pi (10)^2}$$

$$\frac{L_V 10^2}{L_{V0} D^2} = 100^{\frac{1}{5}(M_{V0} - M_V)} *$$

$$\log_{10}\left(\frac{L_v}{L_{v0}}\right) + 2 \log\left(\frac{10}{D}\right) = \frac{2}{5} (M_{v0} - m_v)$$

$$\downarrow$$

$$(0.9 \log_{10}(P) + 2.75)$$

Solve for D :

$$* m_{obs} - m_{true} = A_\lambda$$

$$m_{obs} = m_{true} + A_\lambda$$

$$* \rightarrow \frac{I_1}{I_2} = 100^{\frac{1}{5}(m_2 - m_1)}$$

$$m_{true} = m_{obs} - A_\lambda$$

$$= 100^{\frac{1}{5}(m_2 - (m_{true} + A_\lambda))}$$

$$= 100^{\frac{1}{5}(m_2 - m_{true} - A_\lambda)}$$

$$I_1 = \frac{L_{true}}{4\pi D^2}$$

$$I_2 = \frac{L_{obs}}{4\pi D^2}$$

$$\frac{L_{true}}{L_{obs}} = 100^{\frac{1}{5}(m_{obs} - m_{true})}$$

$$= 100^{\frac{1}{5}(A_\lambda)}$$

$$L_{true} = L_{obs} 100^{\frac{1}{5}A_\lambda} \leftarrow A_B$$

$$* \frac{I_1}{I_2} = 100^{\frac{1}{5}(m_2 - m_1)}$$

abs. mag:

$$* \left(\frac{\frac{L_{true}}{4\pi (10)^2}}{\left(\frac{L_0}{4\pi 10^2} \right)} \right) = 100^{\frac{1}{5}(M_0 - M_B)}$$

$$* \left(\frac{L_{true}}{L_0} \right) = 100^{\frac{1}{5}(M_0 - M_B)}$$

$$\frac{5}{2} \log\left(\frac{L_{true}}{L_0}\right) = M_0 - M_B$$

$$M_B = M_0 - \frac{5}{2} \log\left(\frac{L_{true}}{L_0}\right)$$

$$* S_* \quad \alpha_{rec} \rightarrow R_1 = \alpha_{rec} R_e$$

$$* [I_e/H] = \log(z/z_0) *$$

$$I(R) = I_0 e^{-R/R_H} \rightarrow \text{Magnitudes:}$$

$$\mu(R) = -2.5 \log(I(R)) *$$

$$\frac{I_1}{I_2} = 100^{\frac{1}{5}(m_2 - m_1)} \quad * \left\{ \begin{array}{l} I_2 = 1 \\ m_2 = 0 \end{array} \right\} *$$

$$\rightarrow I_1 = 100^{\frac{1}{5}(-m_1)} \quad \log(I) = \log(10^{-\frac{2}{5}m_1})$$

$$\log(I) = -\frac{2}{5}m_1 \quad I$$

$$m = -\frac{5}{2} \log(I)$$

$$\mu(R) = -2.5 \log(I)$$

$$I = I_0 e^{-R/R_H}$$

$$\mu(R) = -2.5 \left(\log_{10}(I_0) + \log_{10}(e^{-R/R_H}) \right)$$

• change of base $\log_{10}(e) = \frac{\ln(e)}{\ln(10)}$

$$* \log_x(y) = \frac{\log_z(x)}{\log_z(y)} *$$

$$\mu(R) = -2.5 \left(\log_{10}(I_0) + \frac{\ln(e^{-R/R_H})}{\ln(10)} \right)$$

$$= -2.5 \left(\log_{10}(I_0) - \frac{R}{R_H} \left(\frac{1}{\ln(10)} \right) \right)$$

$$= -2.5 \log_{10}(I_0) + \frac{2.5}{\ln(10)} \left(\frac{R}{R_H} \right)$$

PS 3
(14)

$$1.086$$

$$* \mu(R) = \mu_0 + 1.086 \left(\frac{R}{R_H} \right) *$$

$$(1d) \quad \mu_0 = 21.65 \text{ mag arcsec}^{-2}$$

$$\mu_B = 25 \text{ mag arcsec}^{-2}$$

$$\begin{aligned} \hookrightarrow M_B(\tilde{R}_{25}) &= 25 \\ &= \mu_0 + 1.086 \left(\frac{\tilde{R}_{25}}{R_H} \right) = 25 \Rightarrow \left(\frac{\tilde{R}_{25}}{R_H} \right) \end{aligned}$$

We need luminosity: $M_B - (M_{B,25}) = 2.5 \log \left(\frac{L_{\tilde{R}}}{L_{\text{tot}}} \right)$

$$L_{\text{tot}} = \int_0^{2\pi} d\phi \int_0^{\infty} dR R I(R) \quad \uparrow \quad R^0 = 13.7$$

$$L_{\tilde{R}} = \int_0^{2\pi} d\phi \int_0^{\tilde{R}} dR R I(R) \quad \leftarrow \quad a = -\frac{1}{R_H}$$

$$\begin{aligned} L_{\text{tot}} &= \int_0^{2\pi} d\phi \int_0^{\infty} dR R I_0 e^{-R/R_H} \\ &= 2\pi I_0 \int_0^{\infty} dR R e^{-R/R_H} \end{aligned} \quad \left| \int_0^x \tilde{x} e^{a\tilde{x}} d\tilde{x} = \frac{e^{ax}}{a^2} (ax - 1) \right.$$

$\lim_{x \rightarrow \infty} \rightarrow \infty$

$$L_{\text{tot}} = 2\pi I_0 R_H^2$$

$$L_{\tilde{R}} = 2\pi I_0 R_H^2 \left(1 - e^{-\tilde{R}/R_H} \left(1 + \frac{\tilde{R}}{R_H} \right) \right)$$

$$\frac{L_{\tilde{R}}}{L_{\text{tot}}} = \left(1 - e^{-\tilde{R}/R_H} \left(1 + \frac{\tilde{R}}{R_H} \right) \right)$$

$$M_{B,\text{tot}} = 13.7 + 2.5 \log \left(1 - e^{-\tilde{R}_{25}/R_H} \left(1 + \frac{\tilde{R}_{25}}{R_H} \right) \right)$$

6.5 MF:

- States how galaxies form / factors that hinder / effect gal. formation.

(i) Supernova feedback:

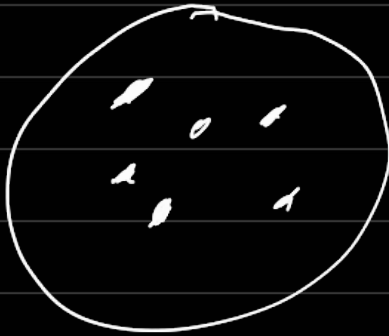
- Some SN: so energetic $\rightarrow$ doesn't allow for gal. formation smaller gal.

$$E_{\text{SN}} > E_{\text{BE}} \leftarrow \text{grav}$$

$\rightarrow$ Reduces the efficiency of formation

(ii) AGNS: Similar Energy output

$$* 2u + \underline{V} = 0$$



$$E_T = K + \underline{U}$$

$$= K - 2K$$

$$E_T = -K$$

QM:

$$E_A = -13.6 \frac{1}{n^2}$$

(Binding E.
Hydrog.)

$$E_{SN} > -\frac{1}{2} M_{gal} \sigma^2$$

$\Psi(M) dM$: The ^{total} mass of gal with a region of M to $M + dM$

$$* \Psi(M) \sim M^{-\beta}$$

$$\ln(\Psi) \sim -\beta \ln(M)$$

→ If there were no feedback. But we have feedback.

feedback: different terms

DM: Does look like a line

$$-\beta e^{-M/M_{*}} \ln(M)$$

